

**Definition****Base**

Let  $(X, T)$  be a topological space. Then  $B \subseteq T$  is a **base** for the topology  $T$  if every open set  $U \in T$  is a union of sets in  $B$ .

We call a set  $B \in B$  **basic open**.

**Proposition**

Suppose  $(X, T)$  is a topological space. Then  $B$  is a base for  $T$  if and only if for every open set  $U \in T$ , for any point  $x \in U$ , there is some basic open set  $B \in B$  such that  $x \in B \subseteq U$ .

If  $B$  satisfies this property, we say it is a base *one point at a time*.

*Proof:* First, suppose  $B$  is a base one point at a time. Then let  $U \in T$  be an arbitrary open set.

For all  $x \in U$ , choose  $B_x$  to be a basic open set with  $x \in B_x \subseteq U$ . We claim

$$U = \bigcup_{x \in U} B_x.$$

First, every point  $x \in U$  is in its own  $B_x$ , so  $U$  is a subset of the union. Furthermore, the union is a union of subsets of  $U$ , so it is also a subset of  $U$ .

Thus  $U$  is equal to the union. Since  $U$  was arbitrary,  $B$  is a base.

Conversely, suppose  $B$  is a base. Let  $U$  be an open set, and let  $x \in U$ . Then  $U$  is a union of basic open sets, meaning

$$U = \bigcup_{i \in I} B_i \quad \text{where } B_i \in B \text{ for all } i \in I.$$

So  $x \in B_j$  for some  $j \in I$ , meaning  $x \in B_j \subseteq U$ . ■

**Defining a Topological Space from a Base**

So far, we have examined cases where  $(X, T)$  is already a topological space with the goal of finding a base for  $T$ . Now, we take  $X$ , not yet a topological space, and we wish to define *open* in a way consistent with the topological axioms.

Let  $B$  be a family of subsets of  $X$  which we want to be *basic open*. One idea is to define  $U$  as *open* if and only if it is a union of basic open sets.

Unfortunately, this does not work. Consider  $X = \mathbb{R}$  without the standard topology, and suppose  $(1, 2)$  and  $\mathbb{Q}$  are our basic open sets. Then  $\mathbb{R}$  is not even open in this space.

We need certain restrictions on which sets are basic open for this idea to work.

### Theorem

Given  $X$  and a family of basic open sets  $B$ , define  $U \subseteq X$  being *open* to mean that  $U$  is a union of basic open sets of  $B$ . Then this space meets the topological axioms if  $B$  satisfies the following properties:

- For every  $x \in X$ , there is a basic open set  $B \in B$  such that  $x \in B$ .
- If  $B_1$  and  $B_2$  are basic open and  $x \in B_1 \cap B_2$ , then there is another basic open set  $B$  where  $x \in B \subseteq B_1 \cap B_2$ .

*Proof:* The first condition is equivalent to  $X$  being open.

Also, the empty set  $\emptyset$  is open because it is a union of the empty family of (exactly zero) open sets.

Suppose  $\bigcup U_i$  is a union of open sets. If  $x \in \bigcup U_i$ , then  $x \in U_j$  for some  $j$ , and we have  $x \in B \subseteq U_j$  for some basic open  $B \in B$ , meaning

$$x \in B \subseteq \bigcup U_i$$

so the entire union is also open.

Suppose  $U_1$  and  $U_2$  are open, and let  $x \in U_1 \cap U_2$ . Then  $x \in U_1$ , so  $x \in B_1 \subseteq U_1$  for some basic open  $B_1 \in B$ . Similarly,  $x \in B_2 \subseteq U_2$  for some  $B_2 \in B$ .

Now  $x \in B_1 \cap B_2$ , so by the second condition on  $B$ , there is some  $B$  such that

$$x \in B \subseteq B_1 \cap B_2 \implies x \in B \subseteq U_1 \cap U_2$$

so  $U_1 \cap U_2$  is also open. ■

### Definition

### Hausdorff

A topological space  $(X, T)$  is **Hausdorff** if for every pair of points  $x$  and  $y$  in  $X$ , if  $x \neq y$ , then there are some disjoint open sets  $U$  and  $V$  for which  $x \in U$  and  $y \in V$ .

In a sense,  $x$  and  $y$  can be *housed off*.